

CHAPTER 1**AgroGuard: A Novel IoT-Enabled Framework for Smart Crop Preservation in Grain Silos**

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Abstract: This paper deals with the design and development of IoT based smart solution to combat the challenges faced by manual monitoring of crops inside silos. A compact hardware model with a dashboard is proposed that constantly updates the information based on the real time readings and calculations from the sensors involved. The Arduino UNO R3 are connected with HCSR04 sensors, DHT11 Sensors and LDR modules are used to obtain the real time data while AI Assistant helps in the decision – making procedure using Artificial Intelligence. This, Agro-Guard is a one stop solution for the farmers who are involved in the process of maintaining crop silos that require intense monitoring and heavily dependent on the environmental conditions like temperature and humidity, as it even helps the farmers understand the next course of action in case of an adverse event.

Keywords: Arduino Uno R3; HCSR04 Sensor; DHT11 Sensor; LDR Module Sensor; React Dashboard; AI Assistant.

INTRODUCTION

Systems based on the Internet of Things (IOT) are highly helpful for tasks that are labor-intensive for humans or require continuous monitoring. This can be attributed to the increasing progress in sensor design and the incorporation of artificial intelligence. In today's busy world, everyone, regardless of their profession tries to squeeze out as much profit as one can from their time.

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This is where Agro-Guard comes in, to help farmers be more productive and maximise profit, while giving them an edge over loss minimisation using very readily available sensors and ground level electronic work, to provide a safe, reliable and cheap way to monitor over silos full of crops. A crop health monitoring system was initially proposed that deals with IOT sensors and addresses resource scarcity in 2024 [1], [2]. Devika T. et al. designed a field monitoring system that was implemented using the Arduino Nano and works on large scale areas and is a cost effective solution [3], Plant growth and health monitoring based on Wireless sensor [4] or IOT based sensors [5] while low cost plant growth and health was proposed by Rahul Narang et al. [6]. Using common IoT protocols, Aman Jaiswal et al. worked on a broad implementation of agricultural health monitoring [7].

A survey of current IoT techniques for crop growth monitoring and quality management was conducted by J. D et al. [8]. All of the works were created with resource scarcity in mind, and they have made use of easily accessible ground level sensors to solve the aforementioned difficulties. Using web applications and temperature sensors, Kirti Tyagi et al. developed a system for crop health monitoring [9].

Ishita Banerjee and P. Madhumathy worked on checking the feasibility of IoT Projects taking into account the financial conditions of the farmer class [10]. This paper plays a major role in the development of further projects in this field as farmers often face economic shortages which may cause them difficulties in buying high end devices which are costly. Nipuna Chamara et al. reviewed the past, present, and future technologies in agricultural IoT [11]. D.K. Sreekantha and Kavya A.M. worked on the general application of IoT based projects in the field of agriculture [12], giving us a wide range of vision on what advancements have been made till date in this field and what gaps still remain unattended. Abdoh Jabbariet al. worked on the analysis of the socio-economic factors and how accepting the farmers are of the IoT based technologies [13] have made works that address a very big underlying issue, the detachment of farmers from technology and the feasibility of such projects given the financial condition of the farmer population, thus directly addressing the socioeconomic effects of the works.

Apart from works performed earlier by the researchers, we have proposed a smart approach to the problem faced by farmers where large silos of grain or crops due to lack of constant monitoring often end up rotting due to various factors like cracks in the silos or humidity rising inside the silo. The proposed design includes an Arduino Uno R3, three different sensors i.e. the HCSR-04 Ultrasonic Sensor, the DHT11 Moisture and Temperature sensor and the LDR (Light

Detecting Resistor) Module sensor. The following section of the paper deals with methodology, its results, discussions and conclusion.

METHODOLOGY

The HCSR04 Ultrasonic sensor is used to shoot out an ultrasonic wave of sound which hits the highest point of the inventory level of the silo and using the time required for the wave to travel back and forth the fill level is calculated as the height if the silo is known. The LDR Module is active low and has a threshold of 400 Lux to detect seepage of light through any cracks in the silo while still accounting for ambient light that may be present in it.

The analog output of the LDR Module is used. The DHT11 sensor measures the air temperature and the air moisture. These are then used to calculate the dew point and the vapor pressure deficit. Thus, these raw data are logged to a .json file and then taken by the ML Model to read and predict based on the training data. The ML Model checks for any anomalies in the sensor readings to understand if a sensor is malfunctioning or if everything is stable. Furthermore, using the Dew Point Value and the Vapor Pressure Deficit value, it determines what kind of risk the silo is facing, or if it is stable. Once the risk is determined, the decision is given on the dashboard as to what procedures to follow to save the crops and time left before the entire silo has to be emptied to prevent a total crop loss.

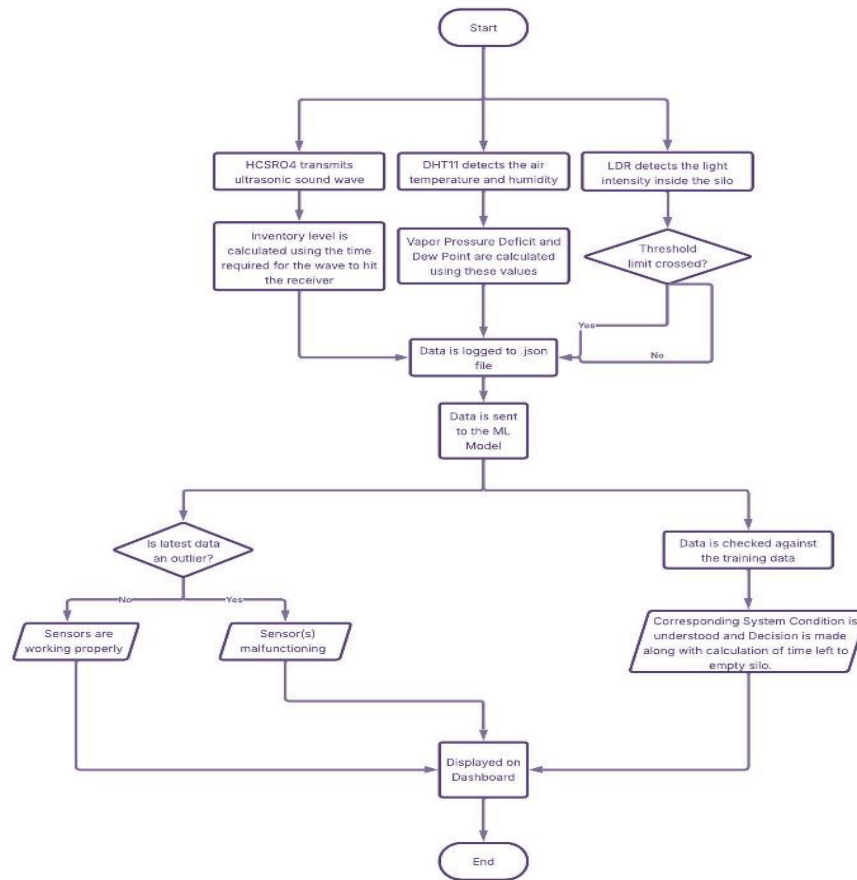


Figure 1: Flowchart showing the working of Agro Guard

The circuit diagram shows the overall connection of the system (as shown in Fig 2).

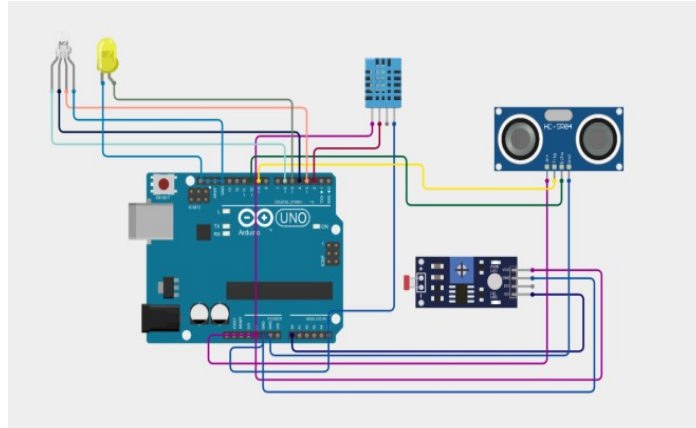


Figure 2: The Circuit Diagram

RESULTS AND DISCUSSIONS

The model metrics is presented in Figure 3.

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--- MODEL METRICS ---
Accuracy: 0.9250
Precision: 0.9773
Recall: 0.8743
F1-Score: 0.9229

Confusion Matrix:
[[1046  23]
 [ 142 988]]

Classification Report:
              precision    recall  f1-score   support

     Bad         0.88         0.98         0.93        1069
     Good         0.98         0.87         0.92        1130

   accuracy              0.93              0.92        2199
  macro avg              0.93              0.92        2199
 weighted avg              0.93              0.92        2199

```

Figure 3: Model Metrics

The classifying algorithm has proven to be very successful, achieving an impressive accuracy rate of 92.5%, along with the impressively high precision of 97.7%. High precision means that there is no doubt in the reliability of the model's judgment when assigning "Good" classification to samples. While recall reaches 87.4%, one can say that the algorithm is inclined to favor safety rather than efficiency: while there are only 23 "Bad" samples that have been erroneously assigned the "Good" status (False Positives), the model seems more cautious with "Good" classifications. In terms of the silo system, it would appear to be a favorable compromise since food safety is prioritized over anything else. The Dashboard displays the data (as shown in Figs 2-7) and the real time temperature, humidity, inventory level and system state. The following images show the versatility and all the states that are recognized by the system.

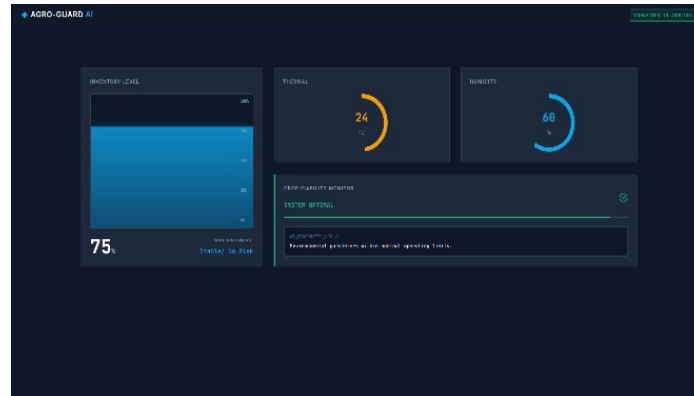


Figure 4: System Optimal

This is the “System Optimal” screen. This comes on the dashboard when the conditions are favorable inside the crop silo. This is the Aridity warning that comes on when the air inside gets very dry and there is a risk of the crops inside shrinking due to lack of moisture. The AI asks to seal the container. This is the Frost warning, which comes on when the system has reached a point where the crops face imminent risk of frost damage. The AI asks the farmer to do thermal intervention to prevent the risk of frosting.

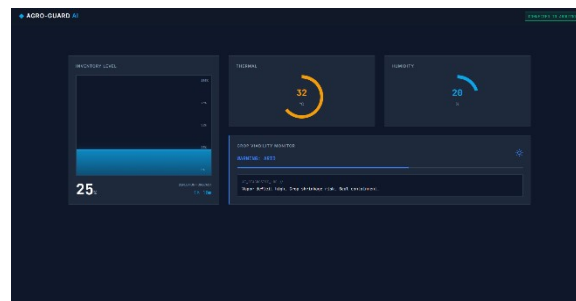


Figure 5: Arid Warning

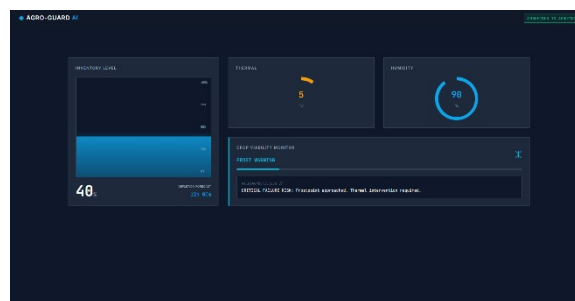


Figure 6: Frost Warning

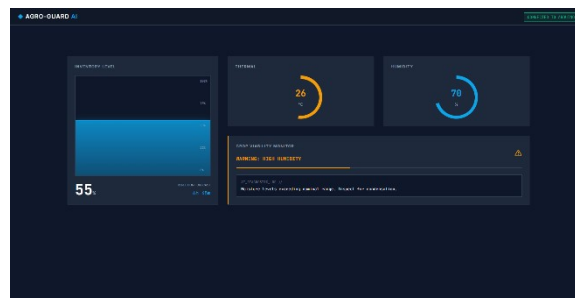


Figure 7: Humidity Warning

This is the Humidity Warning screen comes on when the moisture content inside the silo is reaching alarming levels and can cause biological growth and spoilage of the crop. The AI asks to check for condensation inside the silo.

The Mold Warning comes on when there's high saturation. It usually comes on when the humidity warning is ignored leading to the growth of mold. The AI asks to introduce ventilation.

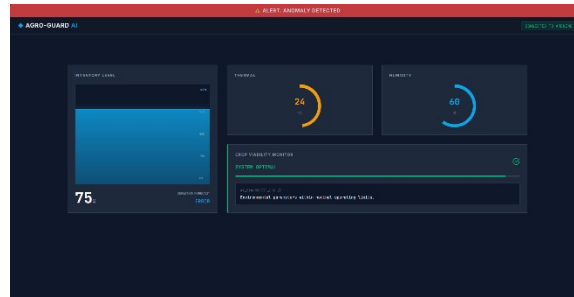


Figure 9: Anomaly Detection Warning

The Anomaly Detection screen comes on when the sensors send a data that is an outlier and confirms something anomalous has happened leading to undesirable reading.

These results show all the necessary data and the AI's decision based on the circumstances.

Table:1: Conditions for proper functioning of the proposed model

Sl No	Message	Condition	Constraint
01	SYSTEM OPTIMAL	$0.8 \leq \text{vpd} \leq 1.5$	$\text{temp} > \text{dew} + 2$
02	WARNING: ARID	$\text{vpd} > 1.5$	$\text{temp} > \text{dew} + 2$
03	FROST WARNING	$\text{temp} \leq \text{dew} + 2$	NONE
04	WARNING: HIGH HUMIDITY	$0.4 \leq \text{vpd} < 0.8$	$\text{temp} > \text{dew} + 2$
05	CRITICAL: MOLD HAZARD	$\text{vpd} < 0.4$	$\text{temp} > \text{dew} + 2$
06	ALERT: ANOMALY DETECTED	$\text{Z-Score} > 3$	NONE

Table 2: Comparison Analysis of ML Model with Recent advancements in same field.

Comprehensive Architectural and Operational Benchmarking of Agricultural ML Models			
Evaluation Dimension	Agro Guard	Crops Prediction Based on Environmental Factors Using Machine Learning Algorithm	Yield prediction, pest and disease diagnosis, soil fertility mapping, precision irrigation scheduling, and food quality assessment using machine learning and deep learning algorithms
Algorithm Architecture	Random Forest (Ensemble)	RF, SVM, LR, DT	CNN, LSTM, XGBoost
Primary Learning Paradigm	Bootstrap Aggregation (Bagging)	Standalone & Ensemble comparison	Deep Neural Networks & Boosting
Data Ingestion & Type	Continuous IoT Streams (VPD, Temp, Hum)	Static Historical Tabular Data (NPK, pH)	High-Dim Spatial/Temporal (UAV, Satellite)
Overfitting Mitigation	Inherent (via tree averaging)	Requires cross-validation tracking	Heavy reliance on Data Augmentation
Deployment & Hardware	Edge-Native (Raspberry Pi/Arduino)	Standard PC / Cloud	Cloud-dependent (Requires GPUs)
Inference Latency	Ultra-low (Millisecond range)	Moderate (Batch processing)	High (Computationally heavy)
Model Interpretability	High (Explicit Feature Importance)	Moderate to High (Depends on model)	Low (The 'Black Box' problem)
Primary Application	Real-time Spoilage & Safety Alerts	Pre-season Crop Suitability	Broad Scale Yield & Pest Diagnosis

CONCLUSION

Agro Guard aims at making the extremely boring and tiring job of maintaining and monitoring over crop silos very easy by automating the job completely and helps the farmers understand what their next step should be to prevent an undesirable event such as full crop spoilage. It has been exclusively trained for Indian agriculture, more specifically in plains, but allows full customization to fully account for any of the land types. Our project has some notable future scopes, which include the implementation of wireless sensors for data transmission instead of wired sensors. We also plan on replacing the Arduino Board with a Raspberry Pi 4B, to make the computing smoother and harness the power of wireless sensor usability. The challenge that needs to be addressed here is the increase in cost as wireless sensors on average are costlier than wired sensors. Moreover, we plan on upgrading the simple dashboard to a full app that directly sends alerts via notifications on mobiles devices (on both Android and iOS). Futhermore we plan

on increasing the accuracy of the project by the use of Time of Flight Sensor (VL53L1X) which eliminates the “dead-zone” issue of HCSR04, ensuring more accurate and versatile results.

CONFLICT OF INTEREST

None declared.

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None declared.

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